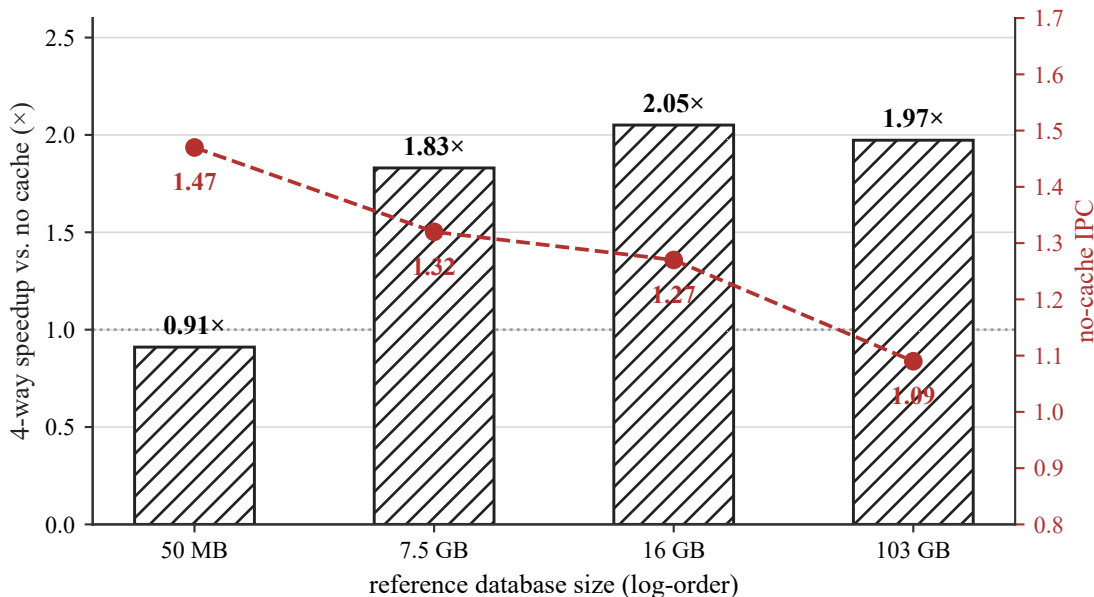


Cache speedup grows as the uncached lookup stalls more



50-read workload, all values from the verified fair batch. Bars: real-hardware-equivalent speedup (cycles / frequency) for the SOFTWARE S2 cache at 4-way. Dashed line: no-cache IPC, falling as the database grows - the mechanism the cache is compensating for. Real Luna HARDWARE cache is fixed throughout: L1d 48KB/12-way, L1i 32KB/8-way, L2 2MB/16-way, L3 105MB/15-way, 2.1GHz.